ENGINEERS WALLAH

CHEMISTRY

Lecture IV

Compounds and Writing Chemical Formulae

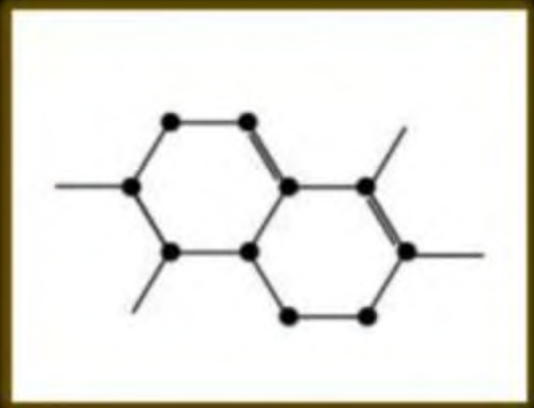


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TOPICS *to be covered*

1. Compounds and Their Relation With Molecules
2. Types of Elements and Compounds
3. Writing Chemical Formulae

Knowledge Ride On



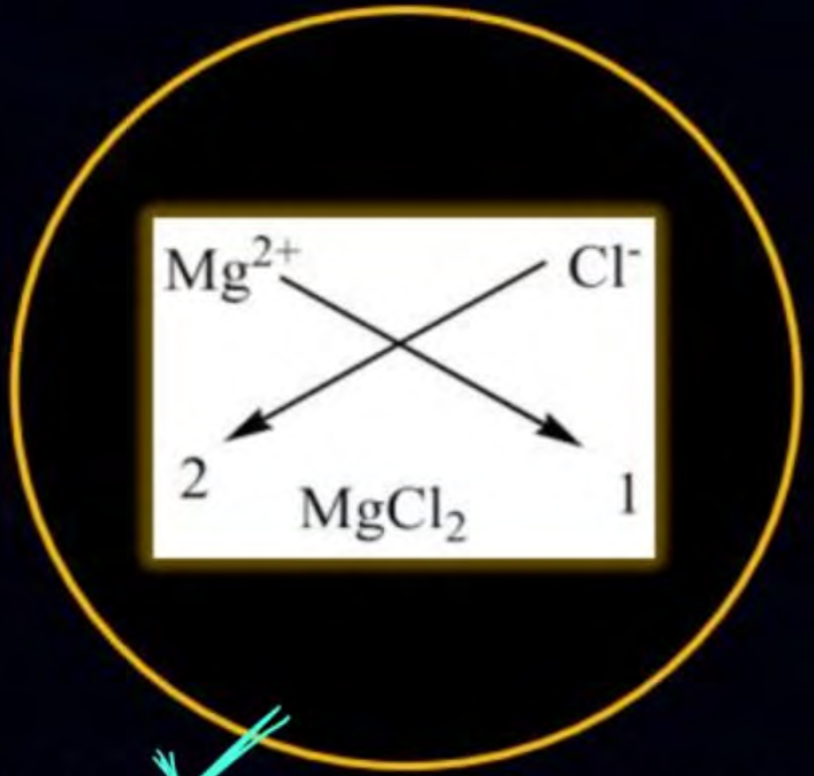
Compounds and Their Relation With Molecules

Knowledge Ride On



Types of Elements and Compounds

Knowledge Ride On



Writing Chemical Formulae

Knowledge Ride On



Efficiency Hacks



Glossary

Compound (यौगिक)
Molecule (अणु)
Cation (धनायन)
Anion (ऋणायन)
Element (तत्व)
Covalent Compound (सहसंयोजक यौगिक)
Ionic Compound (आयनिक यौगिक)

Noble gas (उत्कृष्ट गैस)
Metalloid (उपधातु)
Non-metal (अधातु)
Metal (धातु)

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Atoms of which of the following elements has more than one electron in its outermost shell?

SSC JE 2021

(valence shell)

	Sodium
B	Beryllium
C	Hydrogen
D	Lithium

A.No(z)	K	L	M
11	2	8	1
4	2	2	
1	1		
3	2	1	

Question

The ion (CATION) of an element has 3 positive charges and contains 10 electrons. What will be the atomic number? Also, identify the element.

A. 13 (Aluminium)

B. 10

C. 14

D. 16

X^{3+} → 3 protons more than no. of electrons

⇒ 10 electrons

⇒ 13 protons

Z = 13 ← Aluminium (Al)

Concept Bites – Compounds and Their Relation with Molecules

Compound

A compound is a 'pure substance' made of 'different types of elements, i.e. different types of atoms.' PART II

Part I

Pure substance

↓

Same kind of particles throughout the substance (atoms | ions | molecules)



PART II

Element A



Same type of atoms

+

Element B



Same type of atoms

→

Compound C



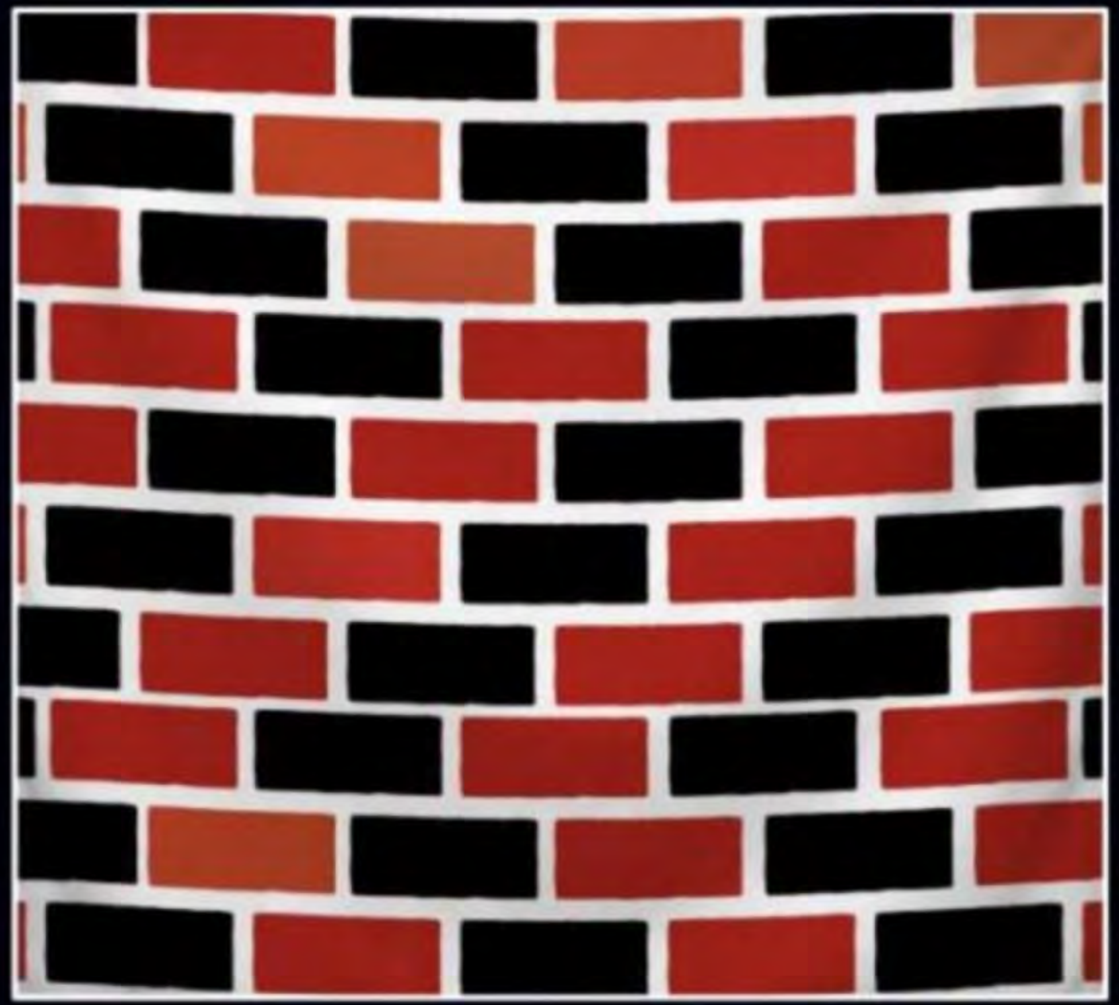
different elements or different atoms

Analogy

Red brick :- One type of atoms

Black brick- Another type of atoms

Wall: COMPOUND



Consider a brick as an atom and there are two bricks, i.e. red and black which combine to form a wall and here it represent a compound.

Concept Bites – Types of Elements and Compounds

Types of Elements

- **Metals** (धातु)
- Non-metals
- Metalloids
- Noble Gases

Some common metals are:
Lithium (Li), Sodium (Na), Potassium (K), Magnesium (Mg), Calcium (Ca), Aluminium (Al), Copper (Cu), Zinc (Zn), Silver (Ag), Gold (Au) etc.

→ Mercury (Hg) is the only metal found in liquid state at room temp. (25 °C).
(Silvery grey)

Types of Elements

- Metals
- Non-metals (अधातु)
- Metalloids
- Noble Gases

Non-metals are:

Solid State

(C) Carbon, (P) Phosphorus, (S) Sulphur, (Se) Selenium, (I) Iodine, (H) Hydrogen, (N) Nitrogen, (O) Oxygen, (F) Fluorine, (Cl) Chlorine (Gaseous State), Bromine (Br)

- Bromine → only non-metal found in liquid state at room temp.
- Bromine → reddish-brown liquid

Types of Elements

- Metals
- Non-metals
- **Metalloids** (उपधातु) — Some properties of metals & some of non-metals
- Noble Gases

Common recognised metalloids are:

Boron (B), Silicon (Si), Germanium (Ge), Antimony (Sb), Arsenic (As), Tellurium (Te), Polonium (Po), Astatine (At)

BAAG Sunil aur Teepu

- B → Boron
- A → Antimony
- A → Arsenic
- G → Germanium
- Sunil → Silicon
- Teepu → Tellurium

Polonium (Po), Astatine (At) → Less recognised as metalloids



Types of Elements

- Metals
- Non-metals
- Metalloids
- **Noble Gases** (उत्कृष्ट गैस)

Noble gases are:

Helium, Neon, Argon, Krypton, Xenon and Radon

(He) (Ne) (Ar) (Kr) (Xe) (Rn)

Practice and PYQ's Wallah



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Question

Which among the following is a non-metal?
(अधातु)

SSC JE 2020

A Potassium (K)

B Aluminium (Al) → Metal

C Magnesium (Mg)

D Nitrogen (N) → Non-metal

Types of Compound

Ionic Compounds
(आयनिक यौगिक)

Consists of → Ions

- Cation
- Anion

ex: $Na^{1+}Cl^{1-}$

Made from → Metal and Non-metal

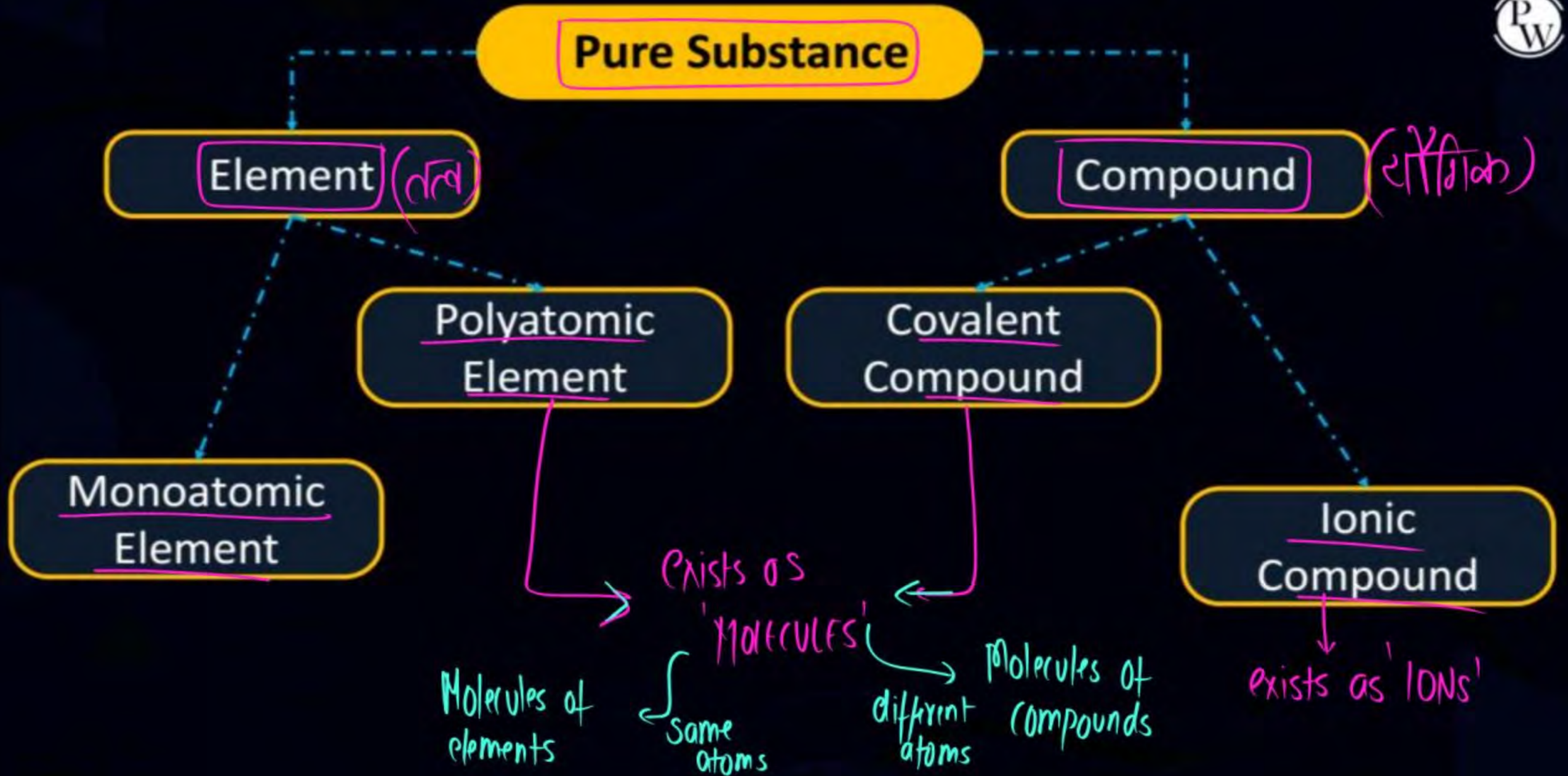
except ammonium (NH_4^{1+})

सहसंयोजक यौगिक
Covalent Compounds

Molecules

Non-metals

ex: H_2O


Pure Substance
Element (तत्व)
Compound (यौगिक)
Polyatomic Element
Covalent Compound
Monoatomic Element
Ionic Compound
Exists as 'Molecules'
Molecules of elements
Same atoms
different atoms
Molecules of compounds
exists as 'IONs'

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Identify the one from the given options that is a compound and consists of molecules?

		Compound (यौगिक)	Molecules of compound (Covalent compound)
A	Chlorine gas (Cl_2) (Element)	×	×
B	Water (H_2O)	✓	✓
C	Nitrogen gas (N_2) (Element)	×	×
D	Sodium chloride (Na^+Cl^-) → Ionic compound	✓	×

Concept Bites – Writing Chemical Formulae

Covalent Compounds

Examples

Cl — K 2, L 8, M 7

8 − 7 = 1

(i) Hydrogen chloride

Gas

Chemical Symbol → H Cl

Valency → 1 1

H_1Cl_1 OR HCl

$HCl(g) \xrightarrow{\text{dissolves in water}}$ Hydrochloric acid (HCl)

(ii) Hydrogen sulphide

Chemical Symbol → H S

Valency → 1 2

H_2S_1 or H_2S

Examples

C (K 2, L 4) O (K 2, L 6) → 8-6=2

(iii) Carbon dioxide

Chemical Symbol → C O

Valency → C: 4, O: 2 ; 2, 1

C_1O_2 or CO_2

N (K 2, L 5) → 8-5=3

(iv) Ammonia

Chemical Symbol → N H

Valency → N: 3 ; H: 1

NH_3

'Ionic Compound'

Examples

(i) Magnesium chloride

	K	L	M
Mg	2	8	2
Cl	2	8	7

- Chemical Symbol → Mg, Cl
- Charge → 2+, 1-
- Valency → 2, 1

$MgCl_2$

(ii) Calcium oxide

	K	L	M	N
Ca	2	8	8	2
O	2	6		

- Chemical Symbol → Ca, O
- Charge → 2+, 2-
- Valency → 2, 2

CaO

Examples

(iii) Aluminium oxide

Chemical symbol → Al O
Valency → 3 2

Al_2O_3

(iv) Potassium oxide

Chemical symbol → K O
Valency → 1 2

K_2O

(Polyatomic Ions - Ionic Compound)

Examples

(i) Potassium nitrate

	K	L	M	N
K	2	8	8	1

- Chemical Symbol → K NO_3
- Charge → $1+$ $1-$
- Valency → 1 1

KNO_3

(ii) Sodium sulphate

	K	L	M
Na	2	8	1

- Chemical Symbol → Na SO_4
- Charge → $1+$ $2-$
- Valency → 1 2

Na_2SO_4

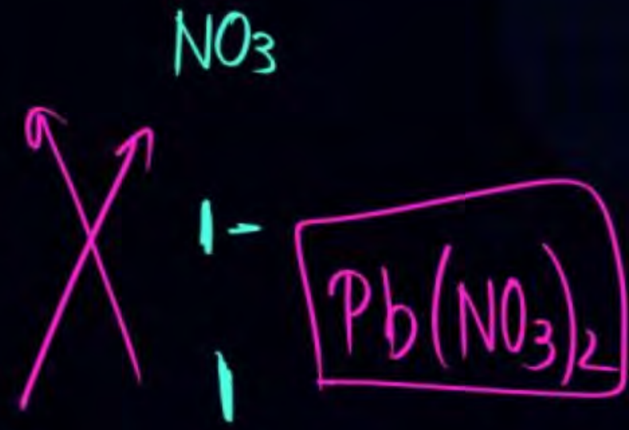
Examples

(iii) Lead nitrate

(Pb)

Pb^{2+} & Pb^{4+}

- Chemical Symbol → Pb
- Charge → 2+
- Valency → 2



(iv) Ammonium hydroxide

- Chemical Symbol → NH_4 OH
- Charge → 1+ 1-
- Valency → 1 1



Examples

(v) Barium sulphate

- Chemical Symbol → Ba SO_4
- Charge → 2+ 2-
- Valency → 2 2

$BaSO_4$

(vi) Sodium bicarbonate

hydrogen carbonate

- Chemical Symbol → Na HCO_3
- Charge → 1+ 1-
- Valency → 1 1

$NaHCO_3$

	Ca		OH
Charge →	2+	X	1-
Valency →	2		1

$Ca(OH)_2$

	Na		CO_3
Charge	1+	X	2-
Valency	1		2

Na_2CO_3

Important to Remember – Part I

Chemical Name	Chemical Formula	Common Name
(i) Solid carbon dioxide	CO_2	(i) Dry ice
(ii) Calcium hydroxide	$Ca(OH)_2$	(ii) Slaked lime
(iii) Sodium hydroxide	NaOH	(iii) Caustic soda
(iv) Potassium hydroxide	KOH	(iv) Caustic potash
(v) Calcium oxide	CaO	(v) Quicklime
(vi) Sodium carbonate	Na_2CO_3	(vi) Soda ash
(vii) Calcium carbonate	$CaCO_3$	(vii) Marble/Chalk/Limestone

Important to Remember – Part I

Chemical Name	Chemical Formula	Common Name
(viii) Sodium chloride	NaCl	(viii) Rock salt/Table salt
(ix) Sodium nitrate	$NaNO_3$	(ix) Chile saltpetre
(x) ~~Nitrogen dioxide~~ Dinitrogen monoxide (Nitrous oxide)	N_2O	(ix) Laughing gas
(xi) Silver nitrate	$AgNO_3$	(x) Lunar caustic

(xii) Potassium nitrate — KNO_3 — (xi) Saltpetre

(xiii) Sodium bicarbonate / Sodium hydrogen carbonate — $NaHCO_3$ — (xii) Baking soda

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

What is the chemical formula of sodium chloride?

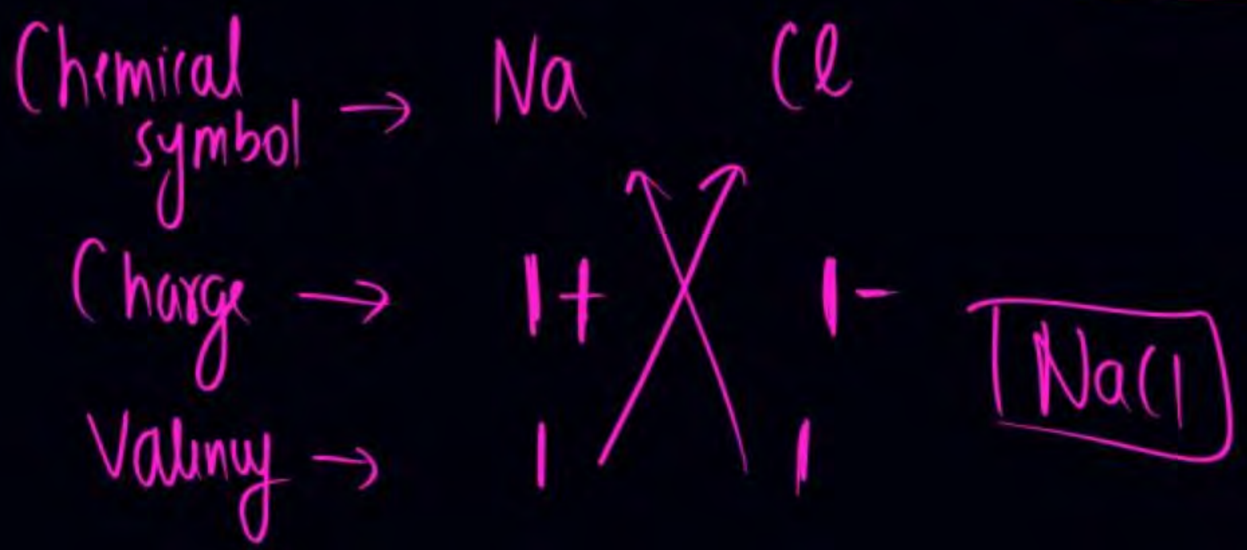
SSC JE 2020

A. $NaCl$

B. Na_2Cl

C. $KMnO_4$

D. $NaCl_2$



Question

Which of the following is known as *'salt peter'*?

RRB JE 2015

A $NaCl$

B KNO_3

C NH_4OH

D $FeSO_4$

Question

The chemical formula $Cu(NO_3)_2$ corresponds to:

RRB JE 2019

A. Copper(II) nitrate

B. Copper nitride

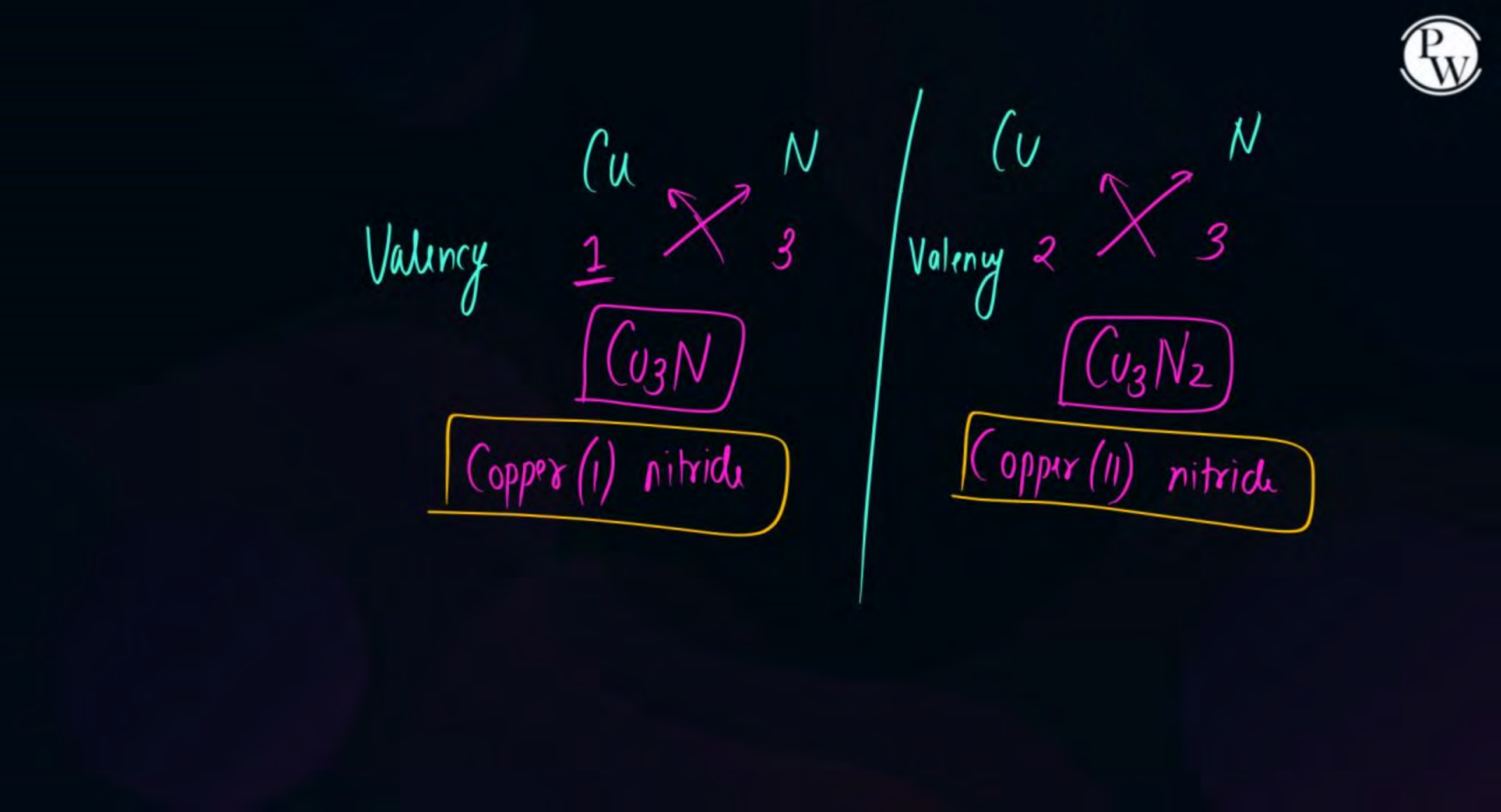
C. Copper(I) nitrate

D. Calcium nitrate

	Cu	NO_3
Charge	2+	1-
Valency	2	1

$Cu(NO_3)_2$

Cu^{1+}	Cu^{2+}


Cu N
Valency 1 3
Cu_3N
Copper (I) nitride
Cu N
Valency 2 3
Cu_3N_2
Copper (II) nitride

Question

What is the common name of sodium bicarbonate?

OR 'Sodium hydrogen carbonate' RRB JE 2015

A ✓ 'Baking soda'

B Salt

C Bleaching powder

D Soda ash

Question

Valence electrons in the element A are 3 and that in element B are 6. Most probable compound formed from A and B is:

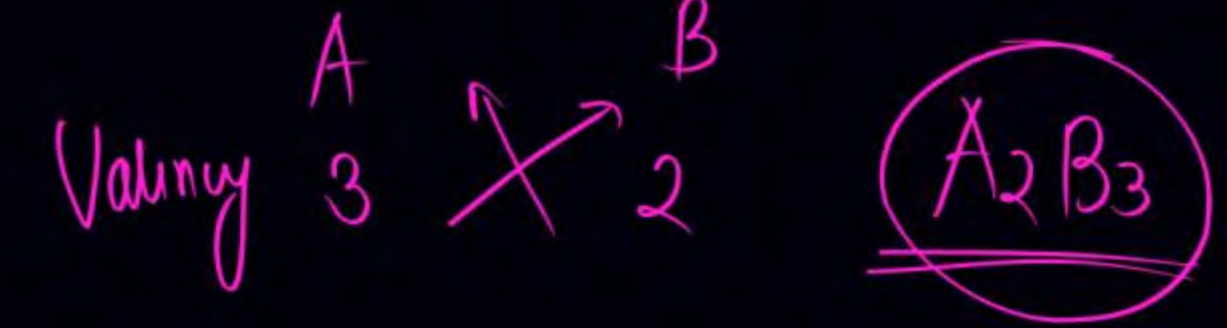
RRB SSE 2014

(A) A_2B

(B) AB_2

(C) A_2B_3

(D) A_3B_2



Question

Which of the following is slaked lime?

SSC JE 2018

A $Ca(OH)_2$

B CaO (Quicklime)

C $CaCO_3$ (Marble/Limestone/Chalk)

D $CaCl_2$

Efficiency Hack!

Foods to Boost Brain Functioning and Memory Power

(i) Walnut, Tuna Fish and Flax Seeds

(ii) Peanuts and Dark Chocolates

(iii) Turmeric




#sbsathhai
#pwsathhai
Thank
You !